

Mineralogical characteristics and soil behaviour of lime-stabilised dredged sediment from Sengguruh Reservoir for sustainable fill applications

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Abstract: Severe sedimentation in the Sengguruh Reservoir has reduced its effective storage capacity to approximately 24% of the original value, creating persistent dredging and spoil bank management problems. This study aimed to evaluate the potential reuse of dredged spoil bank sediment as sustainable fill material through lime stabilisation. Natural sediments and sediments treated with 10% and 15% calcium oxide on a dry weight basis were evaluated experimentally. Laboratory testing included specific gravity, grain-size distribution, Atterberg limits, soil classification based on USCS (unified soil classification system), USDA (United States Department of Agriculture), and AASHTO (American Association of State Highway and Transportation Officials) systems, soil activity analysis, and scanning electron microscopy coupled with energy-dispersive X-ray spectroscopy (SEM–EDX) characterisation to identify mineralogical changes induced by stabilisation. The results showed that lime treatment substantially improved the sediment properties. Specific gravity increased from 2.493 in natural sediment to 2.645 at 15% lime, while the plasticity index decreased from 9.72% to 2.60%. Soil activity also declined – from 2.25 to 0.30, indicating reduced expansiveness and greater material stability. Under AASHTO, soil classification improved from A-5 to A-4, supporting its suitability for fill applications. The SEM–EDX analysis further revealed increased calcium content and reduced the relative abundance of aluminium and silicon, indicating pozzolanic reactions and mineralogical transformation toward more stable soil behaviour. The novelty of this study lies in integrating geotechnical index properties with mineralogical evidence to explain the stabilisation mechanism of reservoir dredged sediment. It demonstrated that 10–15% lime enables its beneficial reuse as a sustainable construction fill material.

Keywords: lime treatment, sediment stabilisation, Sengguruh Reservoir, soil activity, spoil bank sediment

INTRODUCTION

Reservoirs are artificial structures primarily designed to store water by building dams across river systems (Sisinggih *et al.*, 2023; Viari, Masrevanah and Juwono, 2013). Maintaining their long-term functionality requires sustainable management, especially to address factors that diminish their operational capacity (Viari, Masrevanah and Juwono, 2013). One of the most significant challenges affecting reservoirs globally is sedimentation, which involves the ongoing build-up of suspended materials

carried by river flow and deposited on the reservoir bed. If not properly managed, sedimentation can drastically decrease storage capacity and hinder reservoir performance over time (Kumala, 2023; Amitaba *et al.*, 2024).

The Sengguruh Reservoir, situated in the Malang Regency of East Java Province, exemplifies severe impacts of sedimentation. As the first reservoir in the Brantas River system (Brantas River Basin), it protects the downstream Sutami Reservoir from sedimentation and helps generate electrical energy of 2×14.5 MW. However, the Sengguruh Reservoir's effective storage capacity has declined to

only 24% of its original level. Consequently, sediment dredging is necessary throughout the year, although it is constrained by the limited space for spoil bank expansion due to land availability issues (Sartono, Asmaranto and Hendrawan, 2019). This substantial reduction in storage capacity has adversely affected the operational efficiency of the Sengguruh Hydroelectric Power Plant (HEPP), jeopardising both electricity generation and sediment management.

Although various mitigation strategies, such as mechanical dredging and vegetative erosion control, have been implemented, their effectiveness has been less limited. A significant limitation is the restricted availability of spoil bank for storing the annual volume of dredged sediment, which can reach 200,000 m³. The dredged sediment, which is typically stored temporarily in spoil banks, often retains a high moisture content and requires a lengthy drying period before it can be further processed or reused (Sartono, Asmaranto and Hendrawan, 2020).

Given these limitations, repurposing dredged spoil bank sediment as fill material for geotechnical applications, such as embankment reinforcement, offers a promising alternative. The reuse of bottom sediments in a closed-cycle approach has also been proposed as a sustainable strategy to reduce waste accumulation and support the beneficial application of dredged materials (Mazur, Eymontt and Wierzbicki, 2021). However, this approach necessitates a thorough understanding of the physical and mechanical properties of sediments in both their natural and treated forms, since geotechnical parameters provide essential information for evaluating soil behaviour and supporting engineering assessment (Alsharari, Olenko and Abuel-Naga, 2020). Soil stabilisation using additives such as quicklime (CaO) is widely acknowledged for improving soil strength and durability through pozzolanic reactions (Soemitro and Asmaranto, 2015). Lime stabilisation is particularly valued for its availability, cost-effectiveness, and proven success in modifying soil behaviour.

In this context, the study aimed to examine the physical characteristics and classify natural and lime-stabilised spoil bank sediments from the Sengguruh Reservoir to evaluate their suitability as fill materials. This study further explored the mineralogical transformation and activity index of stabilised dredged sediment, aspects that have rarely been examined in the context of reservoir spoil banks. In addition to assessing basic physical parameters, including specific gravity and Atterberg limits (Kaliakin, 2017), the study identified a change in dominant clay mineral type from highly active montmorillonite to low-activity kaolinite following lime stabilisation. These findings improve the understanding of the chemical behaviour and stability potential of lime-treated sandy clay sediments with initially high reactivity.

MATERIALS AND METHODS

STUDY AREA

This study was conducted at the Sengguruh Reservoir, situated at the confluence of the Brantas and Lesti Rivers in Malang Regency, East Java, Indonesia. Commissioned in 1988, the reservoir was designed with a total storage capacity of 21.5 mln m³, which includes an active storage of 2.5 mln m³ and an inactive storage of 19 mln m³ (Perum Jasa Tirta, 2023). However, dredging is

constrained because the annual dredging requirement is at least 300,000 m³ per year, a target that Jasa Tirta I Public Corporation (PJT 1) is unable to consistently achieve (Sartono, Asmaranto and Hendrawan, 2019). The study focused on the spoil bank area next to the reservoir, where dredged sediment is temporarily stored before further processing or removal (Sartono, Asmaranto and Hendrawan, 2020).

MATERIALS AND EQUIPMENT

Materials:

- **dredged sediment samples:** collected from a spoil bank near the Sengguruh Reservoir; these samples represent the natural condition of disposal sediment subjected to laboratory analysis;
- **quicklime (CaO):** used as a lime stabiliser at 10% and 15% content (by dry weight of soil), selected for its availability and effectiveness in enhancing soil strength.

Firoozi and Firoozi (2023) reported that soil treated with 15% lime exhibited the lowest liquid limit/plasticity index (*LL/PI*), whereas untreated soil had the highest *LL/PI* (Firoosi and Firoosi, 2023). In contrast, Utami (2014) conducted laboratory experiments on expansive soil in Pakuwon, Surabaya, and reported that the laboratory California bearing ratio (CBR) test results increased by 5–10% before declining at 15%. Consequently, 10% lime was considered the optimum content for improving the CBR (Utami, 2014). Based on these findings, lime addition rates of 10% and 15% were selected for this study.

Equipment: laboratory testing tools: including pycnometers, hydrometers, sieves, Proctor compaction moulds, Casagrande devices, ovens, balances with 0.001 g accuracy, scanning electron microscope equipped with energy dispersive X-ray spectroscopy (SEM-EDX) and other standard geotechnical equipment.

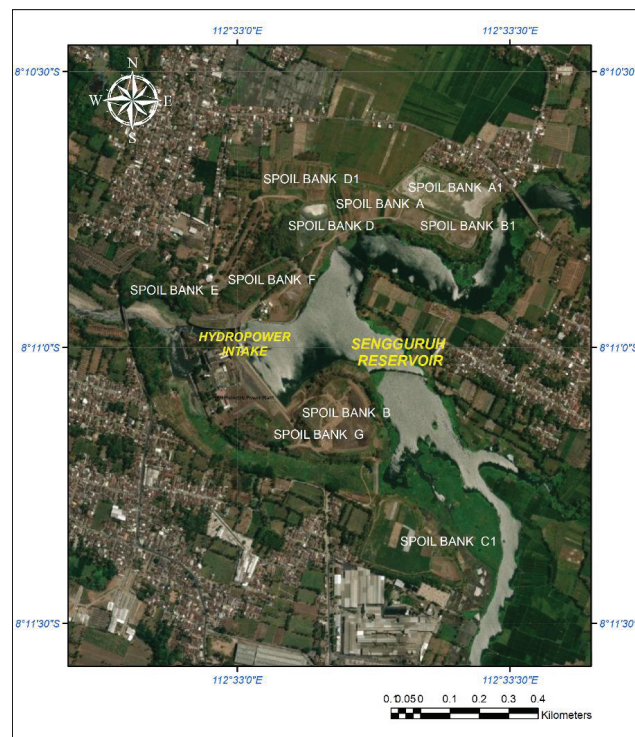


Fig. 1. Research location; source: own elaboration based on: National Geographic, Jasa Tirta I Public Corporation, Survey and Spatial Analyst

RESEARCH METHODS

This research employed a quantitative experimental method, focusing on laboratory testing of physical and mechanical soil properties. The experimental workflow consisted of the following steps.

1. **Sample preparation.** The collected sediment was air-dried and oven-dried to remove moisture. Air-drying was carried out at a room temperature of around 20–30°C, while oven-drying was performed at 110 ±5°C to determine soil water content (ASTM, 2019). After drying, the sediment was sieved for particle-size distribution analysis, and the fraction passing the No. 200 sieve was analysed using a hydrometer in accordance with ASTM D6913-04(2009)e1 (ASTM International, 2009) and ASTM D7928-21e1 (ASTM International, 2021). The dried natural sediment was then mixed thoroughly with quicklime at concentrations of 10% and 15% (by dry weight) to ensure a homogenous mixture prior to testing.
2. **Laboratory testing.** Several tests were conducted to evaluate the physical properties of both natural and stabilised soils, including:
 - **specific gravity (Gs):** determined using ASTM C127-77 (ASTM, 2013) and SNI 15-1990-03 standards (Howard, 1984; ASTM, 2013; Departemen Pekerjaan Umum, 1990);
 - **grain size distribution:** assessed using hydrometer and sieve analysis to determine sand, silt, and clay fractions according to ASTM D6913 and ASTM D7928;
 - **Atterberg limits:** liquid limit (LL), plastic limit (PL), and plasticity index (PI) were evaluated according to ASTM D4318-17e1 (ASTM International, 2017);
3. **Soil classification.** Based on the laboratory data, the sediments were classified using three widely recognised classification systems: AASHTO (American Association of State Highway and Transportation Officials), USDA (United States Department of Agriculture), and USCS (Unified Soil Classification System). Soil activity and mineralogical analysis were also performed.

The soil activity index (*A*) was calculated using Skempton's method (Skempton, 1953), based on the ratio of the plasticity index to the clay content (<0.002 mm). Additionally, SEM-EDX testing was used to examine mineralogical changes and evaluate the transformation of clay types due to lime stabilisation. SEM was employed to observe the microstructure and surface morphology of the samples, while EDX provided qualitative and semi-quantitative information on the distribution of chemical elements within selected areas (Goldstein *et al.*, 2018).

RESULTS AND DISCUSSION

SPECIFIC GRAVITY ANALYSIS

Specific gravity (*G_s*) is a key parameter reflecting the density and solid composition of soil particles. The test results (Tab. 1) indicate a consistent increase in specific gravity values with the addition of lime.

The increase in specific gravity is attributed to the higher density of lime than that of natural sediment particles. Lime stabilisation improves the compactness of the soil matrix by modifying the soil fabric through particle flocculation and agglomeration and, with curing, by promoting pozzolanic

Table 1. Recapitulation specific gravity (*G_s*)

Sample	<i>G_s</i>
Natural sediment (NS)	2.493
NS + 10% lime	2.636
NS + 15% lime	2.645

Source: own study.

reactions. During these reactions, lime reacts with silicates and aluminates in the soil to form cementitious compounds, including calcium-silicate-hydrate (C–S–H) and calcium aluminate hydrate (C–A–H)). These reactions products enhance particle bonding, reduce porosity, and increase the density, strength, and stiffness of the stabilised soil. Similar improvements in strength and microstructural densification following lime-based stabilisation with pozzolanic additives have been reported by James and Pandian (2018). Padmaraj and Arnepalli (2021) also described the formation of pozzolanic reaction products (C–S–H, C–A–H, C–A–S–H) and their role in forming a strong and durable cementation matrix (Sipangkar, Peslinof, and Fendriani, 2023).

GRAIN SIZE DISTRIBUTION

In this study, particle-size distribution was determined using both sieve analysis and the hydrometer method. Sieve analysis was used to determine the distribution of coarse particles such as sand, whereas the hydrometer method was used to determine the distribution of fine particles, including clay and silt (<0.075 mm). The analysis was conducted on three samples.

Particle-size distribution analysis was performed on three samples: natural sediment and sediment stabilised with 10% and 15% lime. The results were used to determine the particle-size composition and classify the sediment. Figure 2 presents the particle-size distribution curves for the three soil samples. The natural sediment exhibited a more gradual curve in the fine-particle (silt-clay) range, indicating the predominance of the silt and clay fractions. Following the addition of 10% and 15% lime, the curve shifts slightly, with an increase in the percentage passing at particle size below 0.01 mm. This change was attributed to the flocculation of fine particles following lime addition Kalkan and Akbulut, 2004). Grain size analysis, conducted using sieve and hydrometer methods, revealed notable changes in particle distribution after lime stabilisation.

The particle-size analysis indicates that the natural sediment, which in this study refers to reservoir sediment obtained through dredging and subsequently deposited in a spoil bank, is dominated by fine sand and silt, comprising 56.94% sand, 38.75% silt, and only 4.31% clay. This particle-size distribution suggests a soil matrix with low clay content and limited natural cohesion, making the particles more susceptible to detachment and transport by surface runoff. Consequently, the material may be associated with a high susceptibility to erosion, especially under rainfall and surface flow conditions. After lime treatment, the particle-size composition changed slightly. The sediment treated with 10% lime comprised 50.69% sand, 41.14% silt, and 8.17% clay, while the mixture with 15% lime comprised 53.64% sand, 37.74% silt, and 8.63% clay. Despite the increase in the measured

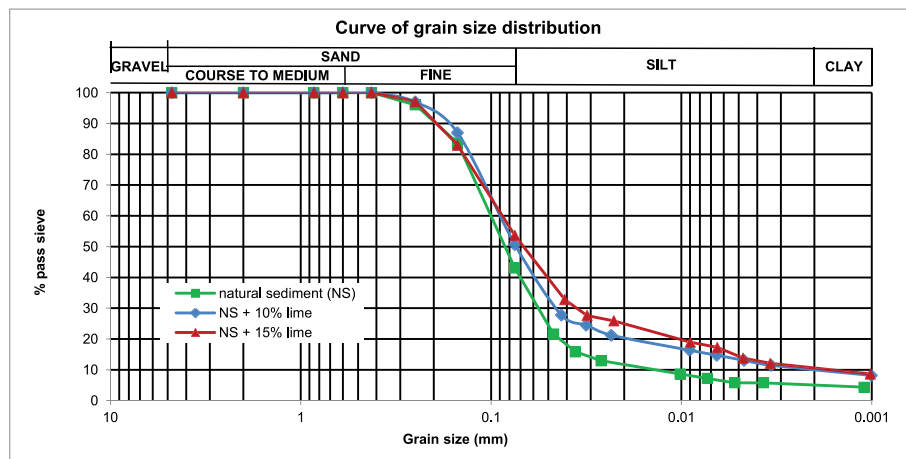


Fig. 2. Soil grain size distribution graph; source: own study

clay fraction following lime addition, the sediment remained predominantly composed of sand and silt, indicating that its overall particle-size distribution can still be associated with a high susceptibility to erosion (Stanek-Tarkowska and Czyż, 2024). Pozzolanic reactions produce calcium silicate hydrate (C–S–H), which acts as a cementitious binding phase and is primarily responsible for increasing the strength and stiffness of the stabilised soil. These reactions promote aggregation of fine particles to aggregate and contribute to into greater soil cohesion and improved engineering behaviour (Padmaraj and Arnepalli, 2021).

ATTERBERG LIMITS AND PLASTICITY INDEX

The water content and plasticity conditions for each sample are shown in Figure 3. The decrease in the plasticity index (PI) with increasing lime content indicates a reduction in soil plasticity and, consequently, improved material stability. The PI decreased from 9.72% to 2.60%, representing a 73.2% decrease, which is consistent with previous findings that lime treatment reduces the mobility of clay particles and increases internal friction (Firoozi and Firoozi, 2023).

SOIL CLASSIFICATION

According to USCS, all samples were classified as ML (silt with low plasticity) whereas under the USDA system, they were classified as sandy clay. According to AASHTO, lime treatment

improved the classification from A-5 (fair to poor) to A-4 (better engineering behaviour), supporting the suitability of the material for its use as a subgrade fill (Howard, 1984).

SOIL ACTIVITY AND CLAY MINERAL TRANSFORMATION

The results of the soil index property tests were used to evaluate the expansion behaviour of the sediment. Skempton (1953) defined soil activity (A) as the ratio of the plasticity index (PI) to the clay fraction (CF) < 0.002 mm and demonstrated that it is related to the mineralogy and geotechnical history of the sediment. In addition, clays are classified as inactive, normal, and active. A nearly linear relationship between PI and CF has been reported for clay deposits, with their ration representing the soil activity (A) (Cerato and Lutenege, 2025).

The substantial reduction in soil activity from 2.25 to 0.30 indicates a marked decrease in the expansiveness of the sediment following lime stabilisation (Fig. 4). According to Skempton (1953), soil activity is closely related to clay mineralogy, with highly active clays typically associated with expansive minerals such as montmorillonite. SEM–EDX analysis showed an increase in calcium content and a reduction in the relative abundance of aluminium and silicon in the stabilised samples, indicating the occurrence of pozzolanic reactions and the formation of cementitious compounds, such as calcium silicate hydrate (C–S–H). Similar studies have reported that lime treatment produces hydrated reaction products, modifies the soil micro-

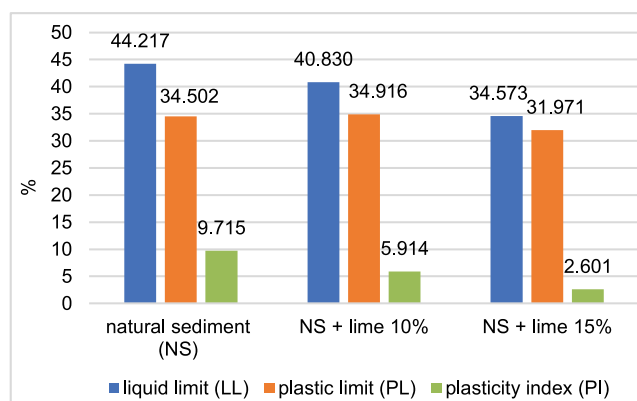


Fig. 3. Atterberg limit test for natural sediment and stabilisation of Sengguruh Dam; source: own study

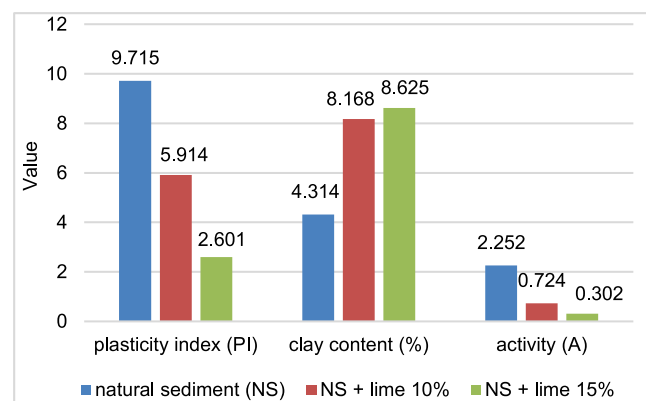


Fig. 4. Activity of disposal sediment and lime stabilised; source: own study

structure, increases strength, and reduces swelling with curing time (Al-Mukhtar, Khattab and Alcover, 2012; Nugroho *et al.*, 2021). The comparative results clearly demonstrate the improvement achieved through lime stabilisation. In particular, the reduction in the plasticity index, the decrease in soil activity, and the improvement in soil classification consistently indicate enhanced engineering properties of the dredged sediment. When considered together with the SEM-EDX observations, these findings suggest that the improved engineering behaviour is associated with mineralogical changes induced by lime stabilisation, representing the principal contribution of this study.

The recapitulation of the distribution and mineralogical content of each soil sample is presented in the Figure 5.

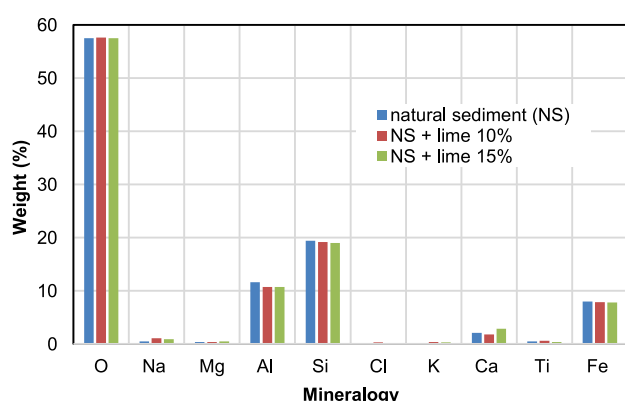


Fig. 5. Mineralogy content: natural sediment disposal of Sengguruh Dam and stabilised; source: own study

SEM-EDX analysis of the Sengguruh Reservoir sediments identified aluminium (Al), silicon (Si), and iron (Fe) as predominant elements. Notably, calcium (Ca) levels increased from 2.1% in the natural sediment (NS) to 2.9% following lime stabilisation, indicating the incorporation of calcium into the stabilised material. This finding is consistent with previous studies reporting that lime addition enhances soil strength and engineering properties through chemical reactions that promote particle bonding and pore filling (Mohd Yunus *et al.*, 2015; Cherian and Arnepalli, 2015). In addition, SEM-EDX analysis showed an apparent increase in sodium (Na) and potassium (K) contents in the stabilised soil. This may be attributed to cation exchange, in which Ca^{2+} supplied by lime displaces Na^+ and K^+ from clay exchange sites into the pore solution, followed by redistribution and local enrichment of these ions within the solid matrix. The interpretation is consistent with previous studies reporting changes in Na and K concentrations during the early stages of ionic exchange in lime-treated soils (Mohd Yunus *et al.*, 2015; Cherian and Arnepalli, 2015). Such ionic exchange between Ca^{2+} , Na^+ , and K^+ is widely recognised as an important short-term mechanism in lime stabilisation before the development of pozzolanic reaction products.

Additionally, chlorine (Cl) was detected solely in the NS treated with 10% lime. this observation may be associated with cation exchange processes similar to those involving potassium and is consistent with previous studies reporting that ionic changes often result from mineral interactions during lime stabilisation (Cherian and Arnepalli, 2015). Iron (Fe) and magnesium (Mg) contents remained relatively unchanged throughout the stabilisation process, indicating that these

elements were less affected by the reactions associated with lime addition, consistent with previous studies describing the selective interaction of lime with certain clay minerals (Mohd Yunus, Marto and Pakir, 2015). Oxygen content also remained relatively stable across all samples, indicating no substantial change in overall composition during stabilisation (Akula, Naik and Little, 2021; Tang *et al.*, 2023).

The reductions in the relative abundance of Si and Al following lime addition may be associated with the formation of hydration compounds during the pozzolanic reaction. Similar changes have been reported in previous studies, which showed that lime stabilisation promotes chemical reactions that modify the soil microstructure and improve its engineering properties (Kalkan and Akbulut, 2004; Cerato and Lutenecker, 2025). Furthermore, soil activity results indicated a reduction in clay activity following lime treatment, suggesting a decrease in the influence of expansive clay minerals. It highlighted a noteworthy transition from montmorillonite in natural sediment to illite and kaolinite types with increased lime content. This transition is consistent with findings that emphasise how lime treatment can diminish soil activity by altering its structural composition, a phenomenon essential for improving soil engineering properties (Goldstein *et al.*, 2018). The shift reflects a reduction in active soil properties, which is beneficial for applications requiring enhanced load-bearing capacities, as corroborated in studies on lime application in soft clay contexts (Al-Mukhtar, Khattab and Alcover, 2012).

In summary, the results indicate that lime stabilisation profoundly improved the mineralogical and mechanical properties of the sediments in the Sengguruh Reservoir. These findings are consistent with previous geotechnical studies demonstrating the importance of lime in soil treatment in enhancing stability and mechanical performance (Howard, 1984; Kalkan and Akbulut, 2004).

The oxygen (O) content remained relatively constant across all samples, reflecting the contribution of oxide-bearing compounds within sediment grains, including SiO_2 , Al_2O_3 , and CaO . The content of Si and Al decreased with increasing lime content, suggesting that some mineral constituents reacted with Ca during lime stabilisation to form hydrate compounds, thereby increasing particle density. Although the Si content decreased following lime addition, it remained higher than that of the other elements, consistent with the predominance of sandy loam in the sediment. This observation agrees with USDA soil classification, which classified both the natural and stabilised sediments as sandy clay.

Scanning electron microscope (SEM) images obtained at 1000× magnification revealed a non-uniform particle structure consisting of fine and coarse grains with rounded and irregular shapes in three different sample types: a) natural sediment, b) natural sediment + 10% lime, and c) natural sediment + 15% lime (Fig. 6).

Dense aggregations were observed in some areas, indicating increased interparticle bonding that is characteristic of clay or silicate materials. The predominance of very fine particles, mainly smaller than 10 μ , indicates a high potential for water absorption and a tendency for expansion and consolidation (Sullivan *et al.*, 2011). This morphology aligns with the sedimentary deposits of the Sengguruh Reservoir, which originate from surface runoff and comprise a mixture of clay, silt, and fine sand. The above morphological characteristics and elemental composition indicate

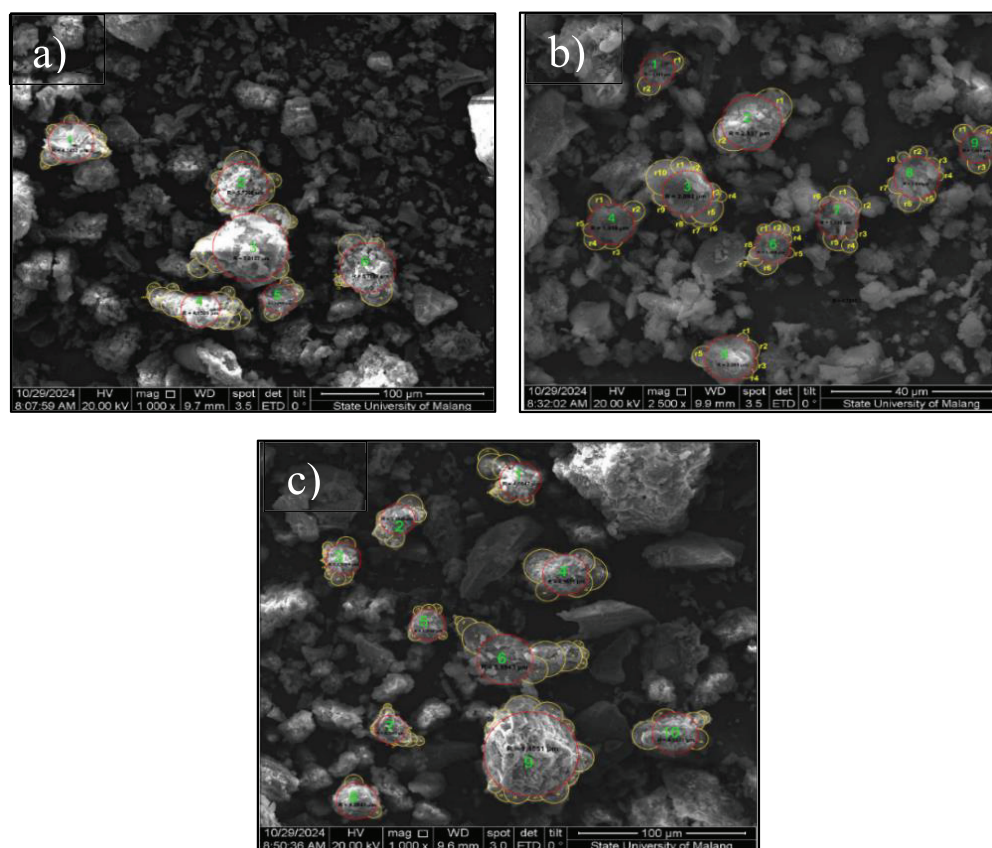


Fig. 6. Scanning electron microscope (SEM) 1000 $\times$ magnification results: a) natural sediment (NS), b) NS + 10% lime, c) NS + 15% lime; source: own study

the presence of Al, Fe, and Si-bearing compounds within the sediment matrix, which are consistent with the fine-grained nature of the material. This type of soil tends to be compressible, with the potential for long-term consolidation if used as a base material or embankment, thus requiring stabilisation. Similar improvement of problematic soils through lime-based stabilisation has been reported by Makinda *et al.* (2018), who showed that lime-sand column treatment can modify the compressibility behaviour of tropical peat. The SEM images also show that the natural sediment (NS) treated with 15% lime exhibited more pronounced particle flocculation and larger aggregates than the NS and NS treated with 10% lime. This reflects the effect of lime stabilisation, in which CaO reacts with soil constituents to form a strong bond between calcium silicate and hydroxide (C-S-D).

Using the Krumbein method, the roundness and sphericity of these grains were quantified, providing critical data for evaluating soil behaviour under load and assessing their potential implications for structural applications (Cherian and Arnepalli, 2015). In the grain-shape analysis, “*a*” denotes the longest principal axis of an individual grain, while “*b*” denotes the intermediate principal axis measured perpendicular to *a*. Therefore, the “*b/a*” ratio represents the proportion of the intermediate axis to the longest grain dimension. Notably, the average *b/a* value of 0.986 indicates that the intermediate and longest axes are nearly equal. The nearly equal intermediate and longest axes, together with the observed roundness and sphericity values, indicate that grain shape may influence interparticle contact, packing density, and shear resistance during compaction, consistent with the findings of Peña, García-Rojo, and Herrmann (2007) on the role of particle shape in sheared dense granular

media. When considered together with the corresponding sphericity and roundness values, this grain-shape characteristic suggests that particle shape may influence friction angle during compaction, an important consideration in the geotechnical design of infrastructure on such soil types (Driss, Harichane and Ghrici, 2022). The sphericity and roundness values determined using the Krumbein method are presented in Table 2.

Based on Table 2, the natural sediment grains have an average sphericity value of 0.8 and an average roundness of 0.4, indicating relatively high sphericity and sub-rounded grain edges according to the Krumbein method. Sphericity represents the degree to which a grain approaches a spherical shape, while roundness describes the smoothness of its edges and corners.

Table 2. Classification of natural sediment grains acc. to Krumbein method

Grain number	Sphericity	Note	Roundness	Note
1	0.9	low	0.5	rounded
2	0.9	low	0.3	sub-angular
3	0.9	high	0.3	sub-angular
4	0.7	low	0.7	rounded
5	0.5	high	0.3	sub-angular
6	0.7	low	0.5	rounded
Mean	0.8	low sphericity	0.4	sub-rounded

Source: own study.

According to the Zingg method, the natural sediment is characterised by a flattened grain form, with an average b/a value of 0.986 and a c/b value of 0.5. Here, a , b , and c represent the longest, intermediate, and shortest principal grain axes, respectively. The high b/a value indicates that the intermediate axis is nearly equal to the longest axis, while the lower c/b value indicates that the shortest axis is considerably smaller than the intermediate axis, confirming the flattened grain morphology. The sphericity and roundness on sediment studied by us is shown in Table 3.

Table 3. Analysis of sphericity and roundness on natural sediment (NS) and stabilised

Method	Shape category	Soil sample		
		NS	NS + 10% lime	NS + 15% lime
Krumbein	sphericity	0.767	0.678	0.720
	category	low sphericity		
	roundness	0.433	0.478	0.400
	category	sub-rounded	rounded	sub-rounded
Measurement	roundness	0.423	0.505	0.455
	category	sub-rounded	rounded	sub-rounded
Zingg	b/a	0.986	1.208	1.011
	c/b	0.500	0.500	0.500
	category	oblate	oblate	oblate

Source: own study.

IMPLICATIONS FOR FILL MATERIAL APPLICATION

The results indicate that lime stabilisation improves the dredged sediment at two complementary levels. At the physical level, lime treatment increases particle bonding and compaction, reduces plasticity, lowers activity, and improves soil classification, indicating better engineering performance. At the mineralogical level, SEM-EDX and microstructural observations confirm an increase in Ca content, reduction in Al and Si proportions, and a more flocculated particle arrangements, reflecting cation exchange and pozzolanic reactions that transform the sediment into a more stable clay material. This dual improvement is important because it not only demonstrates the enhanced suitability of the material for fill applications but also explains the mechanisms responsible for the observed improvement in engineering performance.

The combination of increased specific gravity, reduced plasticity, favourable classification, and mineralogical transformation confirms that lime-treated spoil bank sediment exhibits improved stability and mechanical behaviour, making it a viable candidate for use as fill material. The effectiveness of 15% lime treatment is particularly notable, as it yields low-plasticity soil with kaolinitic characteristics, which are preferred in embankment construction because of their limited volume change and improved shear strength. Although 10% and 15% lime contents were effective for the Sengguruh dredged sediment, these values should not be regarded as universally optimal for all soil types, as the optimum lime content depends on soil mineralogy, plasticity, clay fraction, and the target engineering property.

CONCLUSIONS

This study demonstrates that disposal sediment from the Sengguruh Reservoir, when stabilised with lime, undergoes substantial improvements in its physical and mineralogical properties, making it suitable for reuse as fill material in geotechnical applications. The key findings are as follows. Specific gravity increased from 2.493 (natural) to 2.645 with 15% lime, indicating improved soil density. Soil classification improved from AASHTO A-5 to A-4, with all samples consistently classified as ML (low-plasticity silt) under USCS and Sandy Clay under USDA. The plasticity index decreased by 73.2%, from 9.72% to 2.60%, indicating enhanced soil stability, and the activity index decreased from 2.25 to 0.30, confirming a shift in dominant clay mineral from montmorillonite (highly expansive) to kaolinite (non-expansive), as a result of lime-induced pozzolanic reactions. The addition of 10% and 15% lime to natural sediment did not alter the particle sphericity and roundness, with all particles remaining of low sphericity and sub-rounded to rounded according to the Krumbein criteria. Similarly, based on the Zingg criteria, all particles were classified as oblate. However, during stabilisation, ion exchange occurred between the Ca^{2+} , Na^+ , and K^+ cations, resulting in a more stable soil structure. This process has also been recognised as the primary short-term mechanism in lime stabilisation before the development of pozzolanic products. Ion exchange provides the initial improvement by reducing plasticity and promoting flocculation immediately after lime addition, whereas pozzolanic reactions develop over time, forming stable cementitious compounds that govern the long-term soil stability and durability.

These findings provide strong support for the use of reservoir disposal sediment, which is typically discarded and underutilised, as construction material for river and road embankments, and other geotechnical applications. Although further research is needed, the challenge of sustainable reservoir sediment management can be addressed through an improved understanding of physical, mechanical, and chemical changes associated with lime stabilisation, thereby encouraging the utilisation of reservoir disposal sediment. By quantifying changes in both the macroscopic soil properties and the microscopic mineral structure, this study provides a comprehensive scientific basis for the sustainable use of dredged sediment in civil infrastructure and contributes to broader strategies for sediment management, land optimisation, and the circular use of water and land resources.

CONFLICT OF INTERESTS

All authors declare that they have no conflicts of interest.

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